

Lab 14 –

Exception Handling in Python

Programming Exercise:

Q.1 It is possible to catch multiple (different) exceptions that may be raised in a Try block using multiple (or a series of) except blocks. For example we can write two Except blocks; one for catching an IndexError and the second is for catching ArithmeticError (by taking two lists consisting of equal no of items, and perform division with each pair of list elements). Remember that the IndexError is raised when an element of an array whose index is out of the range of the array is accessed. An out of range index can be either less than zero or greater than or equal to the size of the array.

Source Code:

```
1  # Two lists
2  list1 = [10, 20, 30, 40]
3  list2 = [2, 0, 5, 10]
4
5  try:
6      # Perform division between corresponding elements from the two lists
7      for i in range(len(list1)):
8          try:
9              result = list1[i] / list2[i]
10             print(f"Division of {list1[i]} by {list2[i]} is {result}")
11         except ZeroDivisionError as e:
12             print(f"ArithmeticError: {e}. Can't divide {list1[i]} by zero.")
13         except IndexError as e:
14             print(f"IndexError: {e}. One list is shorter than the other.")
```

```

15 |
16 | # Catch IndexError for the whole operation (if lists are not of the same size)
17 | except IndexError as e:
18 |     print(f"IndexError: {e}. Lists do not have the same length.")
19 |
20 | # Catch any other general exceptions (just in case)
21 | except Exception as e:
22 |     print(f"An error occurred: {e}")
23 |

```

Output:

```

C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\Pychar
Division of 10 by 2 is 5.0
ArithmeticError: division by zero. Can't divide 20 by zero.
Division of 30 by 5 is 6.0
Division of 40 by 10 is 4.0

Process finished with exit code 0

```

Q.2. Design a python program to demonstrate EOFError by using Try-Except Block.

Source Code:

```

1 | # Function to demonstrate EOFError
2 | def handle_eoferror(): 1 usage
3 |     try:
4 |         # Trying to take user input until EOF is encountered
5 |         while True:
6 |             user_input = input("Enter something (Ctrl+D or Ctrl+Z to end): ")
7 |             print(f"You entered: {user_input}")
8 |
9 |         except EOFError:
10 |             print("\nEOFError: No more input available. End of file reached.")
11 |
12 | handle_eoferror()
--

```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\l
Enter something (Ctrl+D or Ctrl+Z to end): Hello
You entered: Hello
Enter something (Ctrl+D or Ctrl+Z to end): Good evening!
You entered: Good evening!
Enter something (Ctrl+D or Ctrl+Z to end): ^D
```

```
EOFError: No more input available. End of file reached.
```

```
Process finished with exit code 0
```

```
|
```

Q.3 Design a python program to demonstrate IndentationError by using Try-Except Block.

Source Code:

```
1  # Function to demonstrate IndentationError
2  def handle_indentationerror(): 1 usage
3      code = """
4  def greet(name):
5  print(f"Hello, {name}!") # Indentation is missing here
6  greet('John')
7      """
8      try:
9          # Trying to execute the code with improper indentation
10         exec(code)
11
12     except IndentationError as e:
13         print(f"IndentationError: {e}. There is an indentation problem in the code.")
14     # Run the program
15     handle_indentationerror()
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\main 3.py"
IndentationError: expected an indented block after function definition on line 2 (<string>, line 3). There is an indentation problem in the code.

Process finished with exit code 0
```

Q.4 Design a python program to demonstrate IOError by using Try-Except Block

Source Code:

```
1  def handle_ioerror(): 1 usage
2      try:
3          # Trying to open a non-existent file for reading
4          with open('non_existent_file.txt', 'r') as file:
5              content = file.read()
6              print(content)
7
8      except IOError as e:
9          print(f"IoError: {e}. The file could not be opened.")
10
11  handle_ioerror()
12
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\main 3.py"
IoError: [Errno 2] No such file or directory: 'non_existent_file.txt'. The file could not be opened.

Process finished with exit code 0
```